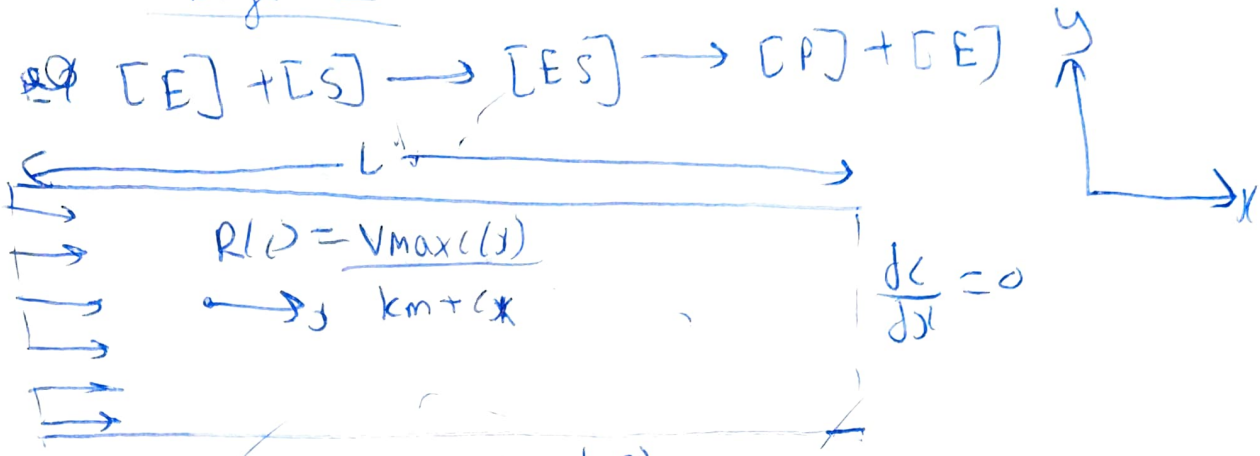
Computational FLUID Dynamics

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Project III

(A)

(a)



c → Substrate Concentration (mol/m³)

Vmax → max possible rate of consumption

km → enzyme rate constant

Given c = c(x)

Enzyme is absorbed/likely present inside the channel

$c(x=0) = c_{in}$

Governing Equation : $\frac{\partial}{\partial t}(c) = D\frac{\partial^2 c}{\partial x^2} - R(c)$ $\quad 0 \le x \le L$

$c(x=0) = c_{in}$

$\frac{\partial c(x=L)}{\partial x} = 0$

u = constant (because of continuity equation)

D → molecular diffusivity of substrate

Non-Dimensionalisation

$c_{ref} = c_{in}$, $u_{ref} = |u|, u = u(u>0)$, $L_{ref} = L$, $x_* = \frac{x}{L_{ref}}$

$c_* = \frac{c}{c_{ref}}$, $u_* = \frac{u}{u_{ref}} = \frac{u}{|u|}$

u > 0 in this case

Substituting these in the governing equation,

$$\frac{\partial (c_* c_{ref} u_* u_{ref})}{\partial (L_{ref} x_*)} = D\frac{\partial^2 (c_{ref} c_*)}{\partial (L_{ref} x_*)^2} - \frac{V_{max} c_* c_{ref}}{k_m + c_* c_{ref}}$$

$$\frac{c_{ref} u_{ref}}{L_{ref}}\frac{\partial (c_* u_*)}{\partial x_*} = \frac{D c_{ref}}{L_{ref}^2}\frac{\partial^2 c_*}{\partial x_*^2} - \frac{V_{max} c_* c_{ref}}{k_m + c_* c_{ref}}$$

Multiplying the entire equation by $\frac{L}{C_{ref} U_{ref}} = \frac{L}{C_{in} U}$

$$\frac{\partial (C_* U_*)}{\partial x_*} = \frac{D}{L_{ref} U_{ref}} \frac{\partial^2 C_*}{\partial x_*^2} - \frac{L_{ref}}{U_{ref} C_{ref}} \left(\frac{V_{max} C_{in} C_*}{K_m + C_{in} C_*} \right)$$

$L_{ref} = L$, $C_{ref} = C_{in}$, $U_{ref} = |U| = U$

$$\frac{\partial (C_* U_*)}{\partial x_*} = \frac{D}{LU} \frac{\partial^2 C_*}{\partial x_*^2} - \frac{L V_{max} C_{in}}{U C_{in}} \left(\frac{C_*}{K_m + C_{in} C_*} \right)$$

$$\frac{\partial (C_* U_*)}{\partial x_*} = \frac{D}{LU} \frac{\partial^2 C_*}{\partial x_*^2} - \frac{L V_{max}}{U} \left(\frac{C_*}{K_m + C_{in} C_*} \right)$$

$$\frac{\partial (C_* U_*)}{\partial x_*} = \frac{D}{LU} \frac{\partial^2 C_*}{\partial x_*^2} - \frac{L V_{max}}{U C_{in}} \left(\frac{C_*}{\frac{K_m}{C_{in}} + C_*} \right)$$

∂x_* comparing with (P3.5)

$\beta = \frac{V_{max} L}{U C_{in}}$, $\alpha = \frac{K_m}{C_{in}}$

$$\therefore \frac{\partial (C_* U_*)}{\partial x_*} = \frac{D}{LU} \frac{\partial^2 C_*}{\partial x_*^2} - \beta \left(\frac{C_*}{\alpha + C_*} \right)$$

$Pe = \frac{U_{ref} L}{D} = \frac{UL}{D}$

$$\therefore \frac{\partial (C_* U_*)}{\partial x_*} = \frac{1}{Pe} \frac{\partial^2 C_*}{\partial x_*^2} - \frac{\beta C_*}{\alpha + C_*}$$

(1) $0 \le x \le L$

$0 \le \frac{x}{L} \le 1$

$0 \le x_* \le 1$

B.C $C(x=0) = C_{in}$

$\therefore \frac{C}{C_{in}}(x=0) = 1$

$\therefore C_*(x_*=0) = 1$

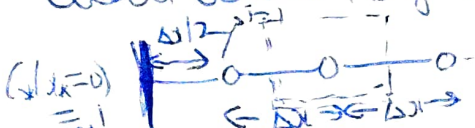
$\left.\frac{\partial C_*}{\partial x_*}\right|_{x_*=1} = 0$

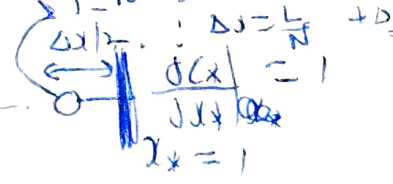
$\therefore \left.\frac{\partial (C_* C_{in})}{\partial (x_* L)}\right|_{x_*=1} = 0$

$\left.\frac{\partial C_*}{\partial x_*}\right|_{x_*=1} = 0$

(b) Mesh Details and approach for discretization

Since it's a 1D problem, we consider a 1D grid.





$L = (N-1)\Delta x + \frac{\Delta x}{2} + \frac{\Delta x}{2}$

$\therefore \Delta x = \frac{L}{N}$

For $2 \le i \le N-1$, where we will use a single governing equation

For $i=1$, Separate governing eqn

For $i=N$, Separate governing eqn.

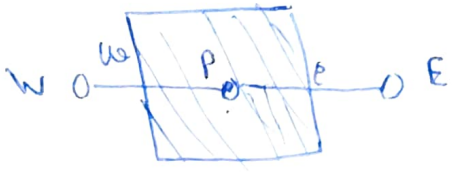
Uniform grid assumption

for all the points, we will use finite volume method.

for upwind scheme, we will approximate value of C_* at phase = value of upstream point.

for central difference scheme, we will take average of both adjust points C_* value. also $i=1 \rightarrow \Delta x/2$, $i=2 \rightarrow \frac{3\Delta x}{2}$

$i=3 \rightarrow \frac{5\Delta x}{2}$, $i=N \rightarrow \frac{\Delta x}{2}+(N-1)\Delta x$



(c) Derivation and Presentation of the final form of the discretized equations along with solution methodology.

for uniform points

$$\int_w^e \frac{\partial}{\partial x}(C_* u_*) = \int_w^e \frac{1}{\rho_e}\frac{\partial^2 C_*}{\partial x_*^2} - \int_w^e \frac{\beta C_*}{\alpha + C_*}$$

$$(C_* u_*)_e - (C_* u_*)_w = \frac{1}{\rho_e}\left(\left(\frac{\partial C_*}{\partial x_*}\right)_e - \left(\frac{\partial C_*}{\partial x_*}\right)_w\right)$$

Since the source term is non-linear, we have to linearize it.

$\left(\frac{-\beta C_*}{\alpha + C_*}\right)$

$$R(C_*) = -\frac{\beta C_*}{\alpha + C_*}$$

$$R(C_*) = -\frac{\beta C_*^{old}}{\alpha + C_*^{old}} + \left.\frac{\partial R}{\partial C_*}\right|_{C_*^{old}} (C_* - C_*^{old})$$

C_* C_*^{old} → guess

$$\frac{\partial R}{\partial C_*} = -\beta\left(\frac{\alpha + C_* - C_*}{(\alpha + C_*)^2}\right) = \frac{-\beta\alpha}{(\alpha + C_*)^2}$$

$$\therefore \left.\frac{\partial R}{\partial C_*}\right|_{C_*^{old}} = \frac{-\beta\alpha}{(\alpha + C_*^{old})^2}$$

$$R(C_*) = \frac{-\beta C_*^{old}}{\alpha + C_*^{old}} - \frac{\beta\alpha}{(\alpha + C_*^{old})^2}(C_* - C_*^{old})$$

$$= \left(\frac{-\beta\alpha}{(\alpha + C_*^{old})^2}\right) C_* - \frac{\beta C_*^{old}}{\alpha + C_*^{old}} + \frac{\beta\alpha C_*^{old}}{(\alpha + C_*^{old})^2}$$

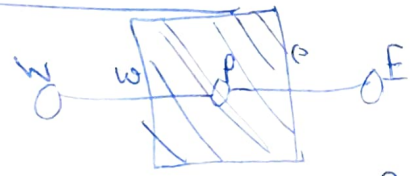
$$= S_C + S_T C_*$$

where $S_C = \frac{-\beta C_*^{old}}{\alpha + C_*^{old}} + \frac{\beta\alpha C_*^{old}}{(\alpha + C_*^{old})^2}$

$$S_T = \frac{-\beta\alpha}{(\alpha + C_*^{old})^2}$$

$$\therefore \frac{\partial}{\partial x_*}(C_* U_*) = \frac{1}{Pe}\frac{\partial^2 C_*}{\partial x_*^2} + (S_C + S_T C_*)$$

For upwind scheme



for cell centered P

$$\int_w^e \frac{\partial}{\partial x_*}(C_* U_*) = \frac{1}{Pe}\int_w^e \frac{\partial^2 C_*}{\partial x_*^2} + \int_w^e (S_C + S_T C_*)\, dx_*$$

Constant profile assumption

$$(C_* U_*)_e - (C_* U_*)_w = \frac{1}{Pe}\left(\left(\frac{\partial C_*}{\partial x_*}\right)_e - \left(\frac{\partial C_*}{\partial x_*}\right)_w\right)$$

$U_{*,e} = 1$, $U_{*,w} = 1$ $(\because U > 0)$ $+ S_{C,P}\Delta x_* + S_{T,P} C_{*,P}\Delta x_*$

$\Delta x_* = \frac{1}{N}$

$\left(\frac{\partial C_*}{\partial x_*}\right)_e$, $\left(\frac{\partial C_*}{\partial x_*}\right)_w$ ⟶ linear profile assumption

$$\therefore (C_* U_*)_e - (C_* U_*)_w = \frac{1}{Pe}\left(\frac{C_{*,E} - C_{*,P}}{\Delta x_*} - \frac{C_{*,P} - C_{*,W}}{\Delta x_*}\right) + S_{C,P}\Delta x_* + S_{T,P} C_{*,P}\Delta x_*$$

$$C_{*,e} = C_{*,E} F_E^{(e)} + C_{*,P} F_P^{(e)}$$

$$C_{*,w} = C_{*,W} F_W^{(w)} + C_{*,P} F_P^{(w)}$$

where $F_W^{(w)} = \max(0, |U_{*,w}|) = 1$, $F_P^{(w)} = \max(0, -|U_{*,w}|) = 0$

$F_P^{(e)} = \max(0, \|u_{*,e}\|) = 1$

$F_E^{(e)} = \max(0, -\|u_{*,e}\|) = 0$

$\therefore$ LHS $= (C_* u_*)_e - (C_* u_*)_w$

$= u_{*,e}\left(C_{*,E} F_E^{(e)} + C_{*,P} F_P^{(e)}\right) - u_{*,w}\left(C_{*,W} F_W^{(w)} + C_{*,P} F_P^{(w)}\right)$

$= A_E^c (C_*)_E + A_P^c (C_*)_P + A_W^c (C_*)_W$

$A_E^c = u_{*,e} F_E^{(e)}$, $A_P^c = u_{*,e} F_P^{(e)} - u_{*,w} F_P^{(w)}$, $A_W^c = -u_{*,w} F_W^{(w)}$

where $u_{*,e} = 1$, $u_{*,w} = 1$

RHS $= \frac{1}{Pe}\left(\frac{C_{*,E} - C_{*,P}}{\Delta x_*} - \frac{C_{*,P} - C_{*,W}}{\Delta x_*}\right) + S_{C,P}\Delta x_* + S_{T,P} C_{*,P} \Delta x_*$

$= A_E^d C_{*,E} + A_P^d C_{*,P} + A_W^d C_{*,W} + Q_P$

$A_E^d = \frac{1}{Pe\,\Delta x_*}$, $A_P^d = -\frac{1}{Pe\,\Delta x_*} - \frac{1}{Pe\,\Delta x_*} + S_{T,P}\Delta x_*$, $A_W^d = \frac{1}{Pe\,\Delta x_*}$

$Q_P = S_{C,P}\Delta x_*$

$A_E^c C_{*,E} + A_P^c C_{*,P} + A_W^c C_{*,W} = A_E^d C_{*,E} + A_P^d C_{*,P} + A_W^d C_{*,W} + Q_P$

$e(C_{*,P})$ $\qquad$ $d(C_{*,P})$

$[A^c][C_*] = [A^d][C_*] + [Q]$

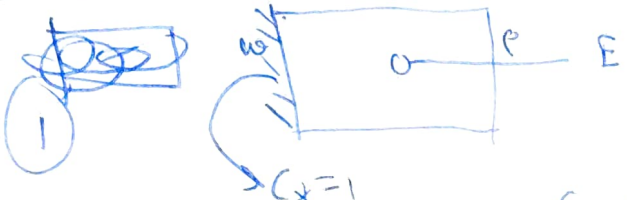
$[A^c - A^d][C_*] = [Q]$

$[C_*] = [A^c - A^d]^{-1}[Q]$

$\rightarrow [A]^{-1}$

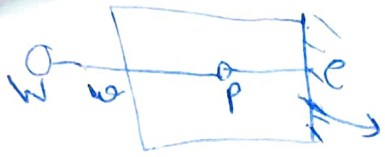
Contrast difference scheme

Boundary conditions



$$LHS = (C_x U_x)_e - (C_x U_x)_w$$
$$= U_{x,e}\left(C_{x,E} F_E^{(e)} + C_{x,P} F_P^{(e)}\right) - U_{x,w}(1)$$
$$e(C_{x,P}) = A_E^c C_{x,E} + A_P^c C_{x,P} + Q_P^c$$
$$A_E^c = U_{x,e} F_E^{(e)}, \quad A_P^c = U_{x,e} F_P^{(e)}, \quad Q_P^c = -U_{x,w}$$
$$\underline{RHS} = \frac{1}{P_e}\left(\left(\frac{\partial C_x}{\partial x_x}\right)_e - \left(\frac{\partial C_x}{\partial x_x}\right)_w\right) + S_{C,P}\Delta x_x + S_{T,P} C_{x,P}\Delta x_x$$
$$= \frac{1}{P_e}\left(\frac{C_{x,E} - C_{x,P}}{\Delta x_x} - \frac{C_{x,P} - C_{x,W}}{\frac{\Delta x_x}{2}}\right) + S_{C,P}\Delta x_x + S_{T,P} C_{x,P}\Delta x_x$$
$$A_E^d = \frac{1}{P_e \Delta x_x}$$
$$e(C_{x,P}) = A_E^d C_{x,E} + A_P^d C_{x,P} + Q_P^d$$
$$A_E^d = \frac{1}{P_e \Delta x_x}, \quad A_P^d = -\frac{1}{P_e \Delta x_x} - \frac{2}{P_e \Delta x_x} + S_{T,P}\Delta x_x = \frac{-3}{P_e \Delta x_x} + S_{T,P}\Delta x_x$$
$$Q_P^d = \frac{2}{\Delta x_x P_e} + S_{C,P}\Delta x_x$$
$$\therefore A_E^c C_{x,E} + A_P^c C_{xP} + Q_P^c = A_E^d C_{x,E} + A_P^d C_{x,P} + Q_P^d$$
$$(A_E^c - A_E^d) C_{x,E} + (A_P^c - A_P^d) C_{x,P} = Q_P^d - Q_P^c$$

(2)



$\rightarrow \frac{dC_*}{dx_*} = 0$

LHS $= (C_* U_*)_e - (C_* U_*)_w$

$= U_{*,e}\left(C_{*,E} F_E^{(e)} + C_{*,P} F_P^{(e)}\right) - U_{*,w}\left(C_{*,W} F_W^{(w)} + C_{*,P} F_P^{(w)}\right)$

$C(C_{*,P}) = A_E^c C_{*,E} + A_P^c C_{*,P} + A_W^c C_{*,W}$

$A_P^c = U_{*,e} F_P^{(e)} - U_{*,w} F_P^{(w)}$, $A_W^c = -U_{*,w} F_W^{(w)}$

RHS $= \frac{1}{Pe}\left(0, -\frac{C_{*,P} - C_{*,W}}{\Delta x_*}\right) + S_{C,P}\Delta x_* + S_{T,P} C_{*,P} \Delta x_*$

$= \frac{1}{Pe}$, $C_{*,P}$

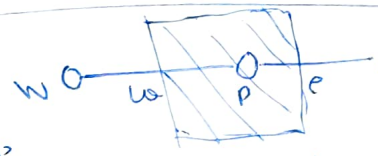
$= A_P^d C_{*,P} + A_W^d C_{*,W} + \Phi_P^d$

$A_P^d = \frac{-1}{Pe\Delta x_*}$, $A_W^d = \frac{1}{Pe\Delta x_*}$, $\Phi_P^d = S_{C,P}\Delta x_*$

$+ S_{T,P}\Delta x_*$ $+ A_P^c C_{*,P} + A_W^c C_{*,W}$

$= A_P^d C_{*,P} + A_W^d C_{*,W} + \Phi_P^d$

Central Difference Scheme (CDS)



for cell centered at P

$\int_w^e \frac{d}{dx}(C_* U_*) = \frac{1}{Pe}\int_w^e \frac{d^2 C_*}{dx_*^2} + \int_w^e (S_C + S_T C_*)\, dx_*$ (constant profile assump.)

$(C_* U_*)_P - (C_* U_*)_w = \frac{1}{Pe}\left(\left(\frac{dC_*}{dx_*}\right)_e - \left(\frac{dC_*}{dx_*}\right)_w\right) + S_{C,P}\Delta x_* + S_{T,P} C_* \Delta x_*$

$(U_*)_e = 1$, $U_{*,w} = 1$

$\left(\frac{dC_*}{dx_*}\right)_e$, $\left(\frac{dC_*}{dx_*}\right)_w$ → linear profile assumption

$C_{*,e} = \frac{C_{*,E} + C_{*,P}}{2} = C_{*,E} F_E^{(P)} + C_{*,P} F_P^{(P)}$

where $F_E^{(P)} = \frac{1}{2}$, $F_P^{(P)} = \frac{1}{2}$

$C_{*,w} = \frac{C_{*,W} + C_{*,P}}{2} = C_{*,W} F_W^{(w)} + C_{*,P} F_P^{(w)}$

where $F_W^{(w)} = \frac{1}{2} = F_P^{(w)}$

LHS $= u_{x,e}\left(c_{x,E} \phi_E^{(e)} + c_{x,P} \phi_P^{(e)}\right) - u_{x,w}\left(c_{x,W} \phi_W^{(w)} + c_{x,P} \phi_P^{(w)}\right)$

LHS $= (c_x u_x)_e - (c_x u_x)_w$

$= u_{x,e}\left(c_{x,E} F_E^{(e)} + c_{x,P} F_P^{(e)}\right) - u_{x,w}\left(c_{x,W} F_W^{(w)} + c_{x,P} F_P^{(w)}\right)$

RHS $= \frac{1}{\rho_e}\left(\left(\frac{\partial c_x}{\partial x_x}\right)_e - \left(\frac{\partial c_x}{\partial x_x}\right)_w\right) + S_{c,P}\Delta x_x + S_{T,P} c_{x,P} \Delta x_x$

$= \frac{1}{\rho_e}\left(\left(\frac{c_{x,E} - c_{x,P}}{\Delta x_x}\right) - \left(\frac{c_{x,P} - c_{x,W}}{\Delta x_x}\right)\right) + S_{c,P}\Delta x_x + S_{T,P} c_{x,P} \Delta x_x$

$= A^d_E c_{x,E} + A^d_P c_{x,P} + A^d_W c_{x,W} + Q_P$

$A^d_E = \frac{1}{\rho_P \Delta x_x}$, $A^d_P = \frac{-2}{\rho_e \Delta x_x} + S_{T,P}\Delta x_x$, $A^d_W = \frac{1}{\rho_e \Delta x_x}$, $Q_P = S_{c,P}\Delta x_x$

LHS $= A^c_E c_{x,E} + A^c_P c_{x,P} + A^c_W c_{x,W}$

$A^c_E = u_{x,e} F_E^{(e)}$, $A^c_P = u_{x,e} F_P^{(e)} - u_{x,w} F_P^{(w)}$, $A^c_W = -u_{x,w} F_W^{(w)}$

$\therefore A^c_E c_{x,E} + A^c_P c_{x,P} + A^c_W c_{x,W} = A^d_E c_{x,E} + A^d_P c_{x,P} + A^d_W c_{x,W} + Q_P$

$e(c_{x,P})$ $\qquad$ $d(c_{x,P})$

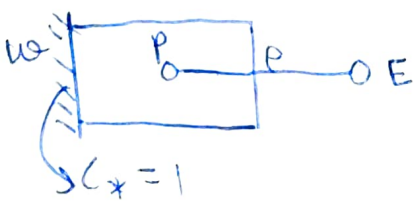
$\therefore [A^c][c_x] = [A^d][c_x] + [Q]$

$[A^c - A^d][c_x] = [Q]$

$\therefore [c_x] = [A^c - A^d]^{-1}[Q]$

Boundary Conditions

(1)



LHS $= (c_x u_x)_e - (c_x u_x)_w$

$= u_{x,e}\left(c_{x,E} F_E^{(e)} + c_{x,P} F_P^{(e)}\right) - (u_{x,w})(1)$

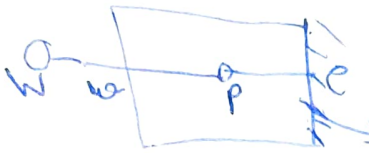
where $F_E^{(e)} = \frac{1}{2} = F_P^{(e)}$

$e(c_{x,P}) = A^c_E c_{x,E} + A^c_P c_{x,P} + Q^c_P$

$A^c_E = u_{x,e} F_E^{(e)}$, $A^c_P = u_{x,e} F_P^{(e)}$, $Q^c_P = -u_{x,w}$

$$\ldots - \frac{1}{Pe}\left(\left(\frac{\partial C_*}{\partial x_*}\right)_e - \left(\frac{\partial C_*}{\partial x_*}\right)_w\right) + S_{C,P}\Delta x_* + S_{T,P}C_{*,P}\Delta x_*$$

(2)



$$\rightarrow \frac{\partial C_*}{\partial x_*} = 0$$

$$\underline{LHS} = (C_* U_*)_e - (C_* U_*)_w$$

$$= U_{*,e}\left(C_{*,E}F_E^{(e)} + C_{*,P}F_P^{(e)}\right) - U_{*,w}\left(C_{*,W}F_W^{(w)} + C_{*,P}F_P^{(w)}\right)$$

$$e(C_{*,P}) = A_E^C C_{*,E} + A_P^C C_{*,P} + A_W^C C_{*,W}$$

$$A_P^C = U_{*,e}F_P^{(e)} - U_{*,w}F_P^{(w)}, \quad A_W^C = -U_{*,w}F_W^{(w)}$$

$$\underline{RHS} = \frac{1}{Pe}\left(0 - \frac{C_{*,P} - C_{*,W}}{\Delta x_*}\right) + S_{C,P}\Delta x_* + S_{T,P}C_{*,P}\Delta x_*$$

$$= \frac{1}{Pe}\frac{C_{*,P} - C_{*,W}}{\Delta x_*}$$

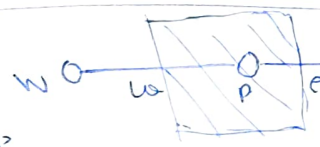
$$= A_P^d C_{*,P} + A_W^d C_{*,W} + \Phi_P^d$$

$$A_P^d = -\frac{1}{Pe\Delta x_*}, \quad A_W^d = \frac{1}{Pe\Delta x_*}, \quad \Phi_P^d = S_{C,P}\Delta x_*$$

$$+ S_{T,P}\Delta x_* \qquad + A_P^C C_{*,P} + A_W^C C_{*,W}$$

$$= A_P^d C_{*,P} + A_W^d C_{*,W} + \Phi_P^d$$

Central Difference Scheme (CDS)



for cell centered at P

$$\int_w^e \frac{d}{dx}(C_* U_*) = \frac{1}{Pe}\int_w^e \frac{d^2 C_*}{dx_*^2} + \int_w^e \underbrace{(S_C + S_T C_*)}_{\text{constant profile assump}} dx_*$$

$$(C_* U_*)_e - (C_* U_*)_w = \frac{1}{Pe}\left(\left(\frac{dC_*}{dx_*}\right)_e - \left(\frac{dC_*}{dx_*}\right)_w\right) + S_{C,P}\Delta x_* + S_{T,P}C_*\Delta x_*$$

$(U_*)_e = 1, \quad U_{*,w} = 1$

$\left(\frac{dC_*}{dx_*}\right)_e, \left(\frac{dC_*}{dx_*}\right)_w \rightarrow$ Linear profile assumption.

$$C_{*,e} = \frac{C_{*,E} + C_{*,P}}{2} = C_{*,E}F_E^{(P)} + C_{*,P}F_P^{(P)}$$

where $F_E^{(P)} = \frac{1}{2}, \quad F_P^{(P)} = \frac{1}{2}$

$$C_{*,w} = \frac{C_{*,W} + C_{*,P}}{2} = C_{*,W}F_W^{(w)} + C_{*,P}F_P^{(w)}$$

where $F_W^{(w)} = \frac{1}{2} = F_P^{(w)}$

$$RHS = \frac{1}{\rho_e}\left(\left(\frac{\partial C_x}{\partial x_x}\right)_e - \left(\frac{\partial C_x}{\partial x_x}\right)_w\right) + S_{c,P}\Delta x_x + S_{T,P}C_{x,P}\Delta x_x$$

$$= \frac{1}{\rho_e}\left(\frac{C_{x,E} - C_{x,P}}{\Delta x_x} - \frac{C_{x,P} - 1}{\frac{\Delta x_x}{2}}\right) + S_{c,P}\Delta x_x + S_{T,P}C_{x,P}\Delta x_x$$

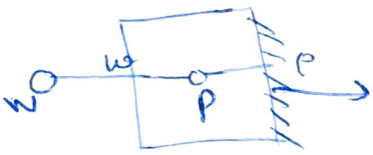
$$a(C_{x,P}) = A^d_E C_{x,E} + A^d_P C_{x,P} + Q^d_P$$

$$A^d_E = \frac{1}{\rho_e \Delta x_x}, \quad A^d_P = \frac{-3}{\rho_e \Delta x_x} + S_{T,P}\Delta x_x, \quad Q^d_P = \frac{2}{\rho_e \Delta x_x} + S_{c,P}\Delta x_x$$

$$A^c_E C_{x,E} + A^c_P C_{x,P} + Q^c_P = A^d_E C_{x,E} + A^d_P C_{x,P} + Q^d_P$$

$$\therefore (A^c_E - A^d_E) C_{x,E} + (A^c_P - A^d_P) C_{x,P} = Q^d_P - Q^c_P$$

(2)



$$\frac{\partial C_x}{\partial x_x} = 0$$

$$LHS = (C_x u_x)_e - (C_x u_x)_w$$

Since, we don't have an E cell for this case, we will have to use upwind scheme for evaluating $C_{x,e}$.

$$\therefore C_{x,e} = C_{x,P}$$

$$\therefore LHS = u_{x,e} C_{x,P} - u_{x,w}\left(C_{x,W} F^{(w)}_W + C_{x,P} F^{(w)}_P\right)$$

where $F^{(w)}_W = F^{(w)}_P = \frac{1}{2}$

$$a(C_{x,P}) = A^c_P C_{x,P} + A^c_W C_{x,W}$$

$$A^c_P = u_{x,e} - u_{x,w} F^{(w)}_P$$

$$A^c_W = -u_{x,w} F^{(w)}_W$$

$$RHS = \frac{1}{\rho_e}\left(0 - \frac{C_{x,P} - C_{x,W}}{\Delta x_x}\right) + S_{c,P}\Delta x_x + S_{T,P}C_{x,P}\Delta x_x$$

$$= A^d_P C_{x,P} + A^d_W C_{x,W} + Q^d_P$$

$$A^d_P = \frac{-1}{\rho_e \Delta x_x}, \quad A^d_W = \frac{1}{\rho_e \Delta x_x}, \quad Q^d_P = S_{c,P}\Delta x_x + S_{T,P}\Delta x_x$$

$$\therefore A^c_P C_{x,P} + A^c_W C_{x,W} = A^d_P C_{x,P} + A^d_W C_{x,W} + Q^d_P$$

So, finally we have derived governing equation for interior cells, both boundary cells, for both upwind and central difference schemes.

We will have coefficient matrix A, solution matrix $[\zeta_*]$

$[A] \rightarrow A_{N\times N}$, $[\zeta_*]_{N\times 1}$, $[Q]_{N\times 1}$.

We have found out all values to populate all three matrices.

(1) $[A][\zeta_*] = [Q]$

$\therefore [\zeta_*] = [A^{-1}][Q]$.

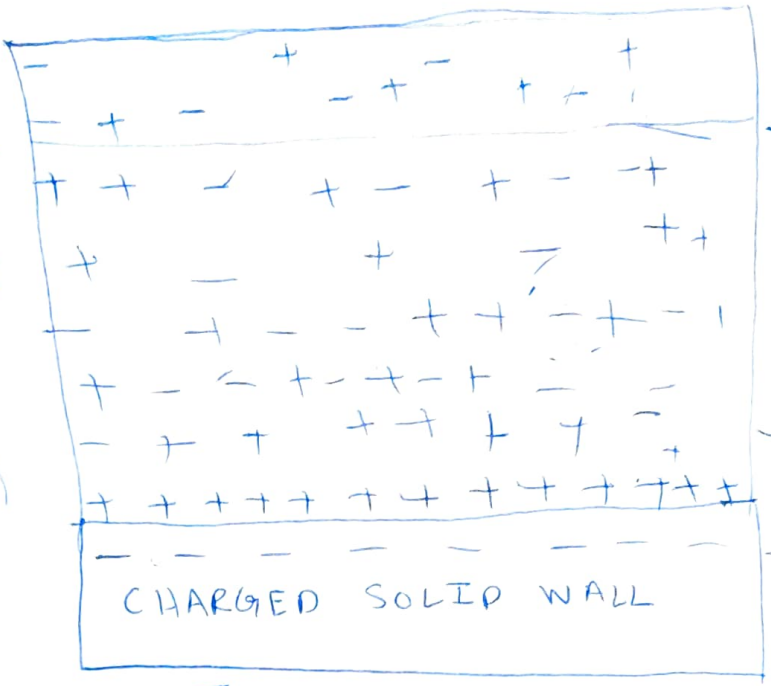
↳ Required Matrix to compute.

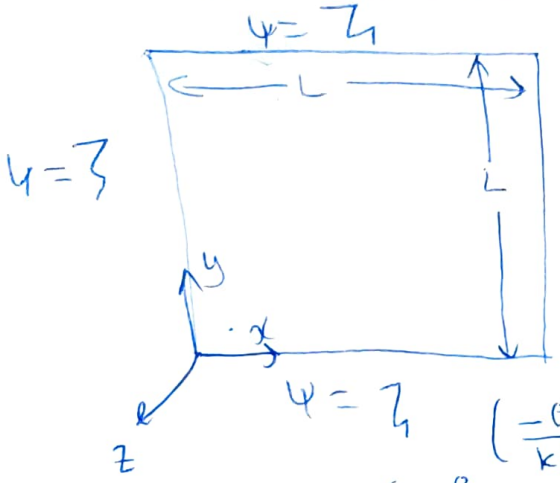
We will vary the values of Pe, α, β to plot ζ_* v/s x_*.

Also the values of four interpolations will be set as per the input Finite difference scheme.

Also, we

B.)





Boundary Conditions

electrostatic potential

Gouien, $C_+ = C_0 e^{\left(\frac{-e\psi}{k_B T}\right)}$, $C_- = C_0 e^{\left(\frac{e\psi}{k_B T}\right)}$

C_+ → positive ion concentration

C_- → negative ion concentration

e → electronic charge

Poisson's Equation : $\nabla^2 \psi = -\rho_f / \epsilon$ (ϵ → permittivity, ρ_f → charge density)

$$\rho_f = e(C_+ - C_-) = e\left(C_0 e^{-\left(\frac{e\psi}{k_B T}\right)} - C_0 e^{\left(\frac{e\psi}{k_B T}\right)}\right)$$

$$= eC_0\left(e^{-\left(\frac{e\psi}{k_B T}\right)} - e^{\left(\frac{e\psi}{k_B T}\right)}\right)$$

$$\rho_f = -2eC_0 \sinh\left(\frac{e\psi}{k_B T}\right)$$

$$\therefore \nabla^2 \psi = -\frac{2eC_0}{\epsilon} \sinh\left(\frac{e\psi}{k_B T}\right) = \boxed{\frac{2eC_0}{\epsilon} \sinh\left(\frac{e\psi}{k_B T}\right)}$$

→ @ P 3.7

Cruiser, the equation is subject to known zeta potential boundary conditions on the flow walls i.e $\psi = \zeta$

$$\nabla^2 \psi = \frac{2ec_0}{\epsilon} \sinh\left(\frac{e\psi}{k_B T}\right)$$

Non-Dimensionalization

$$x_* = \frac{x}{L_{ref}}, \quad y_* = \frac{y}{L_{ref}}, \quad \psi_* = \frac{\psi}{\psi_{ref}}$$

$$L_{ref} = L, \quad \psi_{ref} = \frac{k_B T}{e} \Rightarrow \psi_* = \frac{e\psi}{k_B T}, \quad \zeta_* = \frac{\zeta}{\psi_{ref}} = \frac{\zeta e}{k_B T}$$

Poisson-Boltzman Equation: $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{2ec_0}{\epsilon} \sinh\left(\frac{e\psi}{k_B T}\right)$

Substituting non-dimensional variables into the above equation:

$$\frac{\partial^2 (\psi_* \psi_{ref})}{\partial (x_* L_{ref})^2} + \frac{\partial^2 (\psi_* \psi_{ref})}{\partial (y_* L_{ref})^2} = \frac{2ec_0}{\epsilon} \sinh\left(\frac{e \psi_* \psi_{ref}}{k_B T}\right)$$

$$\therefore \frac{\psi_{ref}}{L_{ref}^2} \frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\psi_{ref}}{L_{ref}^2} \frac{\partial^2 \psi_*}{\partial y_*^2} = \frac{2ec_0}{\epsilon} \sinh\left(\frac{e\psi_*}{k_B T} \times \frac{k_B T}{e}\right)$$

$$\frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\partial^2 \psi_*}{\partial y_*^2} = \frac{2ec_0 L_{ref}^2}{\epsilon \psi_{ref}} \sinh(\psi_*)$$

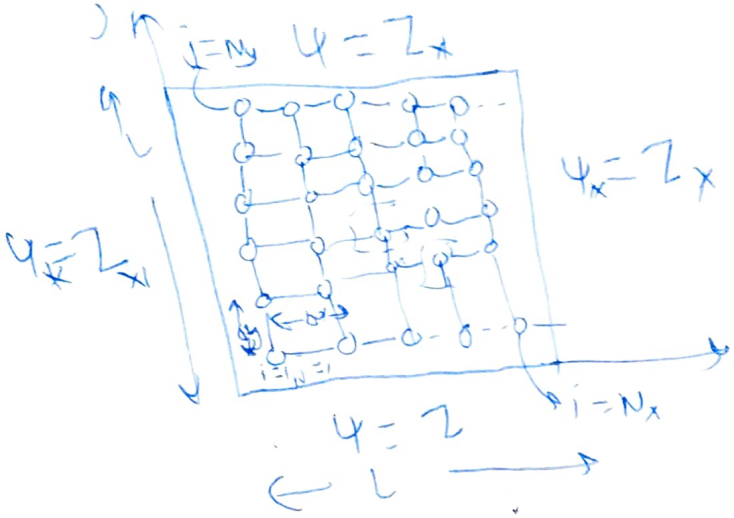
$$\frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\partial^2 \psi_*}{\partial y_*^2} = \frac{2ec_0 L^2 (e)}{\epsilon k_B T} \sinh(\psi_*)$$

$$\therefore \frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\partial^2 \psi_*}{\partial y_*^2} = \frac{2e^2 c_0 L^2}{\epsilon k_B T} \sinh(\psi_*)$$

$\lambda = \sqrt{\frac{2e^2 c_0 L^2}{\epsilon k_B T}}$ → Comparing this with equation P3.9

$$\therefore \frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\partial^2 \psi_*}{\partial y_*^2} = \lambda^2 \sinh(\psi_*)$$

b) Mesh Details and approach for discretization



$(N_x - 1)\Delta x + 2 \times \Delta x/2 = L$

$\therefore \Delta x = \frac{L}{N_x}$

$(N_y - 1)\Delta y + 2 \times \Delta y/2 = L$

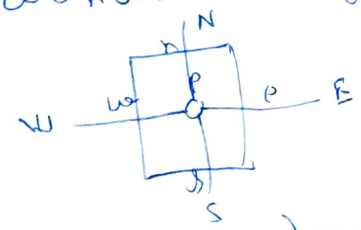
$\therefore \Delta y = \frac{L}{N_y}$

For $i = 2 \rightarrow N_x - 1$, $j = 2 \rightarrow N_y - 1$:

there will be one governing equation

For boundaries there will be one

c) Derivation and final form of equations :



$$\frac{\partial^2 \psi_*}{\partial x_*^2} + \frac{\partial^2 \psi_*}{\partial y_*^2} = \lambda^2 \sinh(\psi_*)$$

$\sinh(\psi_*) \rightarrow$ non linear term

$\therefore \sinh(\psi_*) = \sinh(\psi_*^{old}) + \cosh(\psi_*^{old})(\psi_* - \psi_*^{old})$

$= \sinh(\psi_*^{old}) - \cosh(\psi_*^{old})(\psi_*^{old}) + \cosh(\psi_*^{old})(\psi_*)$

$= S_C + S_T \psi_*$

$$\int_s^n \int_w^e \frac{\partial^2 u_x}{\partial x^2} + \int_s^n \int_w^e \frac{\partial^2 u_x}{\partial y^2} = \int_s^n \int_w^e S_c + S_T u_x$$

$$\left(\frac{\partial u_x}{\partial x}\right)_e - \left(\frac{\partial u_x}{\partial x}\right)_w \Delta y + \left(\left(\frac{\partial u_x}{\partial y}\right)_n - \left(\frac{\partial u_x}{\partial y}\right)_s\right) \Delta x$$

$$= S_c \Delta x \Delta y + S_T u_x \Delta x \Delta y$$

$$\left(\frac{u_{xE} - u_{xP}}{\Delta x} - \frac{u_{xP} - u_{xW}}{\Delta x}\right) \Delta y$$

$$+ \left(\frac{u_{xN} - u_{xP}}{\Delta y} - \frac{u_{xP} - u_{xS}}{\Delta y}\right) \Delta x$$

$$= S_c \Delta x \Delta y + S_T u_x \Delta x \Delta y$$

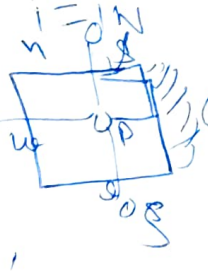
$$A_E u^*_E + A_P u^*_P + A_S u^*_S + A_W u^*_W + A_N u^*_S + \text{①}_P = 0$$

for all $k = (j-1) N_x + i$

$P \to k$, $S \to k - N_x$, $N \to k + N_x$, $E \to k+1$, $W \to k-1$

For left, right, top, bottom boundary

$i = N_x$, $j = N_y$, $j = 1$



left, $\left(\frac{\partial u}{\partial x}\right)_0 = \frac{u_P - u_W}{\Delta x/2}$

rest all terms remain the same

right

Boundary for top $\left(\frac{\partial u}{\partial y}\right)_n = \frac{u_P - u_N}{\Delta y/2}$